Modeling and Forecasting USD/INR Exchange Rate Volatility: An Asymmetric GARCH Approach with Structural Break Analysis

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1. Introduction

1.1 Statement of the Forecasting Problem:

The core objective of this paper is to develop and estimate robust time series models for forecasting the conditional volatility (risk) and mean (return) of the USD/INR exchange rate. This focus is vital because emerging economies like India are central to the modern global financial landscape, making an understanding of their financial complexities highly beneficial for risk management. While forecasting the returns themselves is challenging due to weak serial correlation, the persistence and predictability of their volatility, as noted by Engle (1982), make it an ideal target for analysis. Therefore, this study will incorporate the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) framework to accurately model and forecast the essential currency market risk.

1.2 Motivation and Relevance:

In emerging countries like India, being able to accurately forecast exchange rate volatility is critically important for several stakeholders:

a. **Financial Risk Management**: "A *risk-neutral world* can be defined as an imaginary world where investors require no compensation for bearing risks" (Hull, 2023). Currency volatility represents the primary financial risk for multinational corporations and financial institutions. Precise volatility forecasts are, in fact essential for Value-at-risk(Var), setting hedge ratios and optimizing portfolio allocations (Christoffersen, 2012).
b. **Macroeconomic Stability and Policy Planning:** The exchange rate is a key indicator of India's external stability for banks and policy makers. High and unpredictable volatility can destabilize trade, inflate import costs, and trigger capital flight.
c. **Asset pricing:** Volatility is a fundamental driver of pricing in currency derivatives.

2. Data

2.1. Dataset and Transformation

The dataset used in this study comprises **daily closing prices** for the **USD/INR exchange rate (P_t)** spanning from January 2010 to Sep 2025, providing a total of 4097 observations corresponding to the effective trading days. Financial returns, including the USD/INR series, typically deviate significantly from the standard Normal Distribution, primarily exhibiting heavy , skewness, and time-varying volatility characterized by volatility clustering. The normality JB test conducted on the data confirms this (Figure 1, Appendix 1). Given this confirmed non-normality, all models are estimated using **quasi-Maximum Likelihood Estimation (QMLE)** assuming a Gaussian distribution, which provides consistent standard errors and remains robust to the non-normal distribution of the returns.

Selam Mahmud Ali
48015A

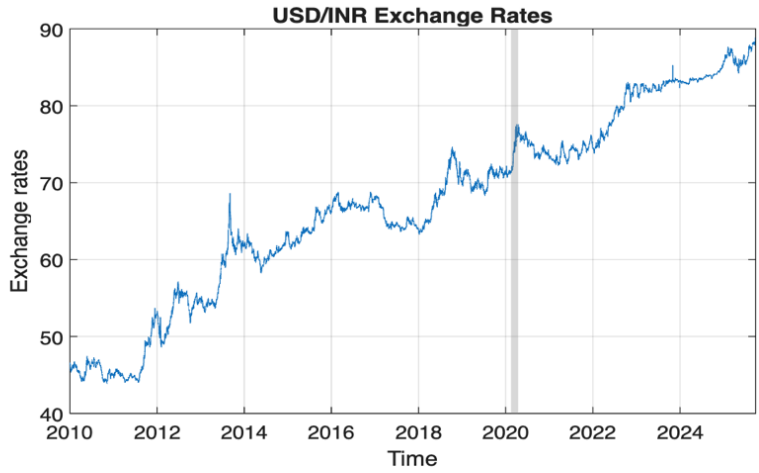

Figure1

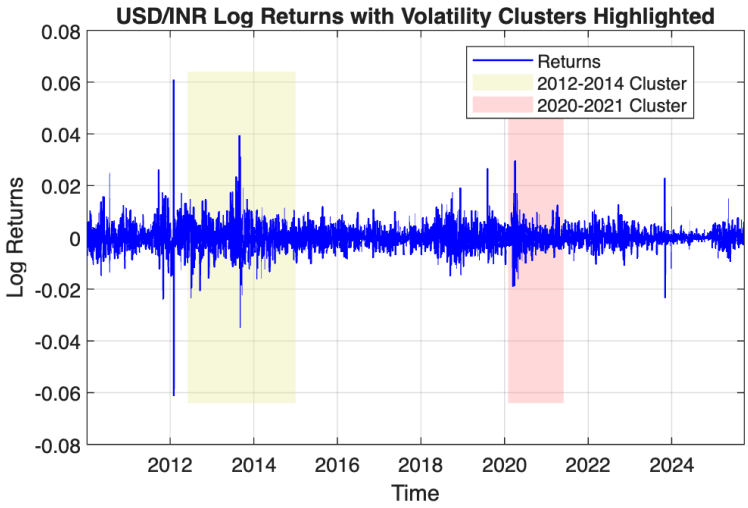

Figure2

The **vertical gray line around 2020** is the most significant structural marker on Figure1. A **structural break** is a sudden change in the underlying data generating process (the trend or volatility).

- The exchange rates appear to be at a relatively high level (around 75) just before this line.
- Immediately following the line, the exchange rates move sharply higher and establish a new, steeper upward trajectory. This suggests a major economic or policy event occurred around 2020 that caused an **acceleration** in the rate of depreciation of the Indian Rupee against the US Dollar.

As is standard practice in financial econometrics, the analysis is conducted on the **logarithmic difference of the Exchange rates** given by the **Returns (R_t)** to ensure a stationary mean process:

$$R_t = \ln(P_t) - \ln(P_{t-1})$$

2.2. Time Series Features and Volatility Clustering

Near-Zero Mean and Low Serial Correlation: The returns fluctuate closely around zero (Figure 2).

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48015A

Prominent Clusters (Figure 2):

- There is a clear high-volatility cluster around **2012-2014**, the so called Taper Tantrum (Bernanke, B. S., 2013);
- Another significant cluster appears around **2020**, likely corresponding to the global shock event (like the COVID-19 pandemic) that was visually marked on the original exchange rate plot.

A volatility cluster is a period where the **magnitude of the returns** (both positive and negative) is much larger than normal, but the **long-run average return remains close to zero**.

The total sample is systematically divided into two distinct periods: an **estimation (in-sample) period** and a **forecasting (out-of-sample) evaluation period**. The estimation period comprises 80% (3276 observations) of the entire time series data, leaving the remaining 821 observations for the evaluation period. This division ensures that the models are tested on genuinely unseen data, meeting the industry standard requirement that the out-of-sample period constitutes at least 20% of the full dataset.

2.3. VIX (CBOE Volatility index) and transformation:

In the Alternative model suggested later on, an external timeseries (for the same timespan as the exchange rates) is included. This external variable is the **VIX**, often referred to as the "Fear Index," which represents the market's 30-day forward-looking expectation of volatility based on S&P 500 options. The VIX serves as a proxy for **global risk aversion and uncertainty**. The VIX timeseries is not used directly but it is included as

$$Z_{VIX,t-1} = \ln\left(\frac{VIX_{t-1}}{VIX_{t-2}}\right)$$

.

3. Methodology:

3.1. Structural Break Analysis

To ensure the mean equation is correctly specified, a formal test for **structural breaks** was performed.
The **Chow Test (Chow, 1960)** was applied to the mean equation:

$$R_t = c + \phi_1 R_{t-1} + \varepsilon_t$$

$$\boxed{H0: \beta_1 = \beta_2 \quad ; \quad H1: \beta_1 \neq}$$

This test determines whether the estimated coefficients are significantly different across sub-samples defined by two major hypothesized economic shocks:

1. **May 22, 2013:** Corresponding to the "Taper Tantrum" announcement(Bernanke, B. S., 2013).

2. **March 16, 2020:** Corresponding to the initial global COVID-19 market shock.

The Chow test compares the residual sum of squares from the single **restricted** regression (using the entire sample) against the combined residual sum of squares from the two separate **unrestricted** regressions (one before and one after the break date). The **rejection of the null hypothesis** (coefficients are equal) for a specific date confirms a statistically significant shift in the mean dynamics.

The outcome of this analysis dictates the inclusion of corresponding **exogenous dummy variables** in the mean equation of the competing models. This accounts for the permanent shifts in the mean return, allowing the volatility models to more accurately capture the time-varying risk.

3.2. Model selection

Following the descriptive analysis and visualization of the returns series, the preliminary orders for the models were determined through an analysis of the correlation structures inherent in the data.

1. **Mean Equation Order Selection (AR):** The **Autocorrelation Function (ACF)** and the **Partial Autocorrelation Function (PACF)** (Figure 2, Appendix 1) of the raw logarithmic returns (Rt) were examined. The PACF showed a significant spike at the first lag, indicating that the returns process is best modeled by a AR(1) structure to begin with.
2. **Variance Equation Order Selection (GARCH):** The ACF and PACF were then applied to the **squared returns (Rt^2)**. The slow decay observed in the ACF of the squared returns (Figure 3), coupled with significant spikes in the PACF at low lags, strongly suggested that the conditional variance exhibits both GARCH and ARCH effects. Based on standard practice for financial returns, a parsimonious specification was selected to model the persistence of volatility.

This preliminary inspection led to the selection of the structure as the basis for the **Benchmark Model**, ensuring that the starting specification is consistent with the empirical properties of the returns.

3.3. Benchmark model

Benchmark Model (ARIMAX(1,0,0)–GARCH(1,1))

Mean equation:

$$R_t = c + \phi_1 R_{t-1} + \beta_1 D_{2020,t} + \varepsilon_t$$

Variance equation:

$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \gamma_1 \sigma_{t-1}^2$$

where $D_{2020,t}$ is a dummy variable indicating the post–March 2020 period.

3.4. Alternative Models

EGARCH:

To complement the symmetric GARCH benchmark, we estimate an Exponential GARCH (EGARCH) model, which allows volatility to respond asymmetrically to return shocks (Nelson, 1991). This framework is particularly suited for testing the **leverage effect**—the notion that negative shocks (i.e., INR depreciation) may increase volatility more than positive shocks.

The **mean equation** remains identical to the benchmark specification:

$$R_t = c + \phi_1 R_{t-1} + \beta_1 D_{2020,t} + \varepsilon_t.$$

$$\ln(\sigma_t^2) = \omega + \sum_{i=1}^{p} \gamma_i \ln(\sigma_{t-i}^2) + \sum_{j=1}^{q} \alpha_j \left(\frac{|\varepsilon_{t-j}|}{\sigma_{t-j}} - E\left[\frac{|\varepsilon_{t-j}|}{\sigma_{t-j}} \right] \right) + \sum_{j=1}^{q} \lambda_j \frac{\varepsilon_{t-j}}{\sigma_{t-j}}$$

Here lambda measures the leverage. If it is >0 then negative shocks have a higher impact on volatility.

Inclusion of the Global Risk Factor (VIX)

Furthermore, an **exogenous time series** is introduced into the conditional variance equation to test the hypothesis of **volatility spillover** from global financial markets to the exchange rate (USD-INR). By including the VIX Log Return in the variance equation, we test whether a surge in global market stress significantly drives an increase in the conditional volatility of the returns. A positive and statistically significant coefficient would confirm that global risk is a **leading indicator** of local currency risk.

Alternative Model (ARIMAX(1,0,0)–EGARCH(1,1) with VIX-in-Variance)

Mean equation:

$$R_t = c + \phi_1 R_{t-1} + \beta_1 D_{2020,t} + \varepsilon_t$$

Variance equation (log form):

$$\ln(\sigma_t^2) = \omega + \gamma_1 \ln(\sigma_{t-1}^2) + \alpha_1 \left(\frac{|\varepsilon_{t-1}|}{\sigma_{t-1}} - E\left[\frac{|\varepsilon_{t-1}|}{\sigma_{t-1}}\right] \right) + \lambda_1 \frac{\varepsilon_{t-1}}{\sigma_{t-1}} + \delta_1 Z_{\mathrm{VIX},t}$$

where $Z_{\mathrm{VIX},t}$ denotes the VIX log return, and λ_1 captures the leverage effect.

3.5. Out-of-Sample Forecast Evaluation

Once all competing models (Benchmark and Alternative) have satisfied the diagnostic checks (Ljung-Box tests for residual adequacy), they are used to generate **one-step-ahead forecasts** over the previously reserved **out-of-sample evaluation period** (observations).

The core objective of this stage is model comparison. The models' ability to predict the **mean return** and the **conditional volatility** in the unobserved sample will be rigorously compared using The Root Mean **Squared Forecast Error for selection of optimal model** .

The Root Sqaured Forecast Error is calculated as:

$$\mathrm{RMSFE} = \sqrt{\frac{1}{H} \sum_{t=T_{\mathrm{est}}+1}^{T} \left(R_t - \hat{R}_{t|t-1} \right)^2}$$

Here's what it means:

- R_t = actual value at time t
- $\hat{R}_{t|t-1}$ = forecasted value of R_t based on information up to $t-1$
- T_{est} = last period used for estimation
- T = last period in your sample
- $H = T - T_{\mathrm{est}}$ = number of forecasted periods

The most **efficient model** will be selected based on a two-step hierarchy:

1. **Forecast Accuracy:** The model with the **lowest RMSE** will initially be favored, as this indicates the highest overall predictive precision for the mean return.
2. **Parsimony:** If two or more models yield statistically or economically **indistinguishable forecast errors** (e.g., their values are nearly identical), the principle of **Parsimony** will govern the final selection. In this scenario, the **simplest adequate model** (the one with the fewest parameters, which corresponds to the model with the lowest **BIC** value) will be chosen as the preferred forecasting model.

4. Results:

4.1. The Chow test reports:

Hypothesized Break Date	Event	Chow F-Statistic	P-Value	Conclusion
May 22, 2013	**Taper Tantrum Announcement**	8.6555	0.0002	**Reject** (Significant Break) : Large F and p-value <significance level
March 16, 2020	**COVID-19 Market Shock**	3.9074	0.0202	**Reject** (Significant Break): Large F and p-value <significance level

Table 1

The rejection of the null hypothesis confirms that both periods introduced a statistically significant shift in the mean parameters of the return series. Given the objective of finding the most parsimonious forecasting model, the structural break dummies will be incorporated into the mean equation of the competing models as exogenous variables and their significance compared.

4.2. Benchmark model

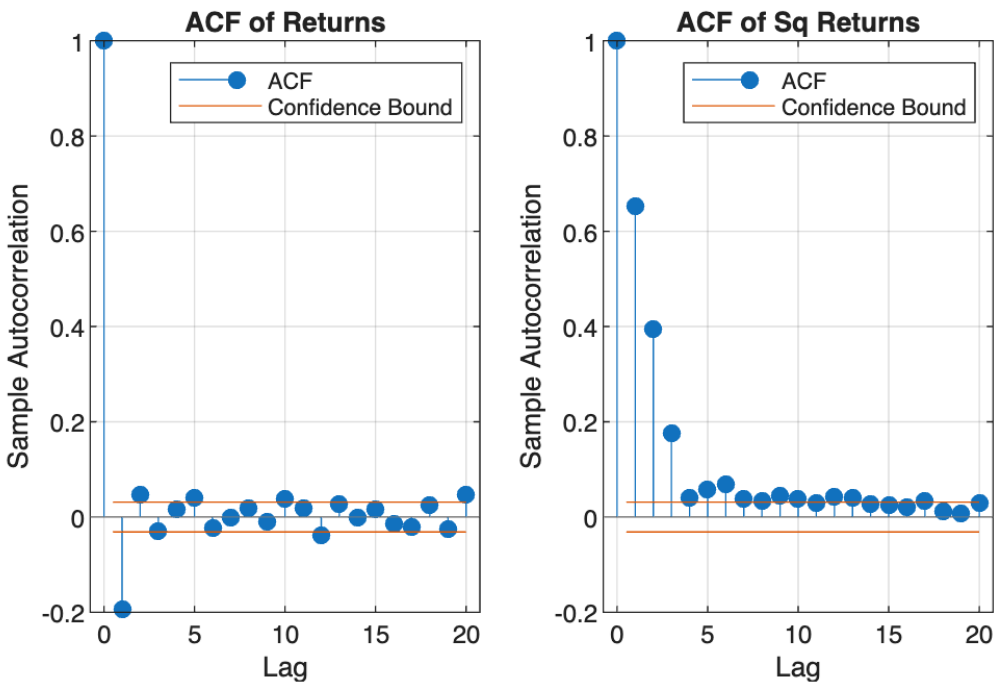

Figure 3

Since both the ACF(Figure3) and the PACF of the returns of Exchange Rates have significant spikes at Lags 1 , and then drop rapidly to zero afterwards, this pattern is ambiguous and suggests a process that might be a combination of Autoregressive and Moving Average components, i.e., an **ARMA(1,1) model.**

ARCH EFFECT:

The correlation in Rt^2 (Figure3), as referred to in the Methodology section, also indicates ARCH effects leading to using a combined ARIMA-GARCH model.

The formal identification of significant structural breaks, specifically, the Taper Tantrum shock in 2013 and the COVID-19 shock in 2020 (as confirmed by the Chow tests), mandates the inclusion of **exogenous dummy variables ($D_{2013} + D_{2020}$)** in the mean equation of the time series models. These dummy variables account for the **statistically significant shift in the expected mean return** that occurred subsequent to each respective event, thereby ensuring the stationarity assumption of the process is maintained and preventing residual bias.

Model Class	Mean Equation Component	Variance Equation Component	LB Test on SR [Mean check]	LB Test on SR^2 [variance check]	ADEQUATE MODEL? p-value>5%
Benchmark(1 dummy)	{AR}(1) + D_{2020}	{GARCH}(1,1)	p=0.0588684	p=0.261954	Yes, Yes
Benchmark (2 dummy)	{AR}(1) $D_{2013} + D_{2020}$	{GARCH}(1,1)	p=0.0561223	p=0.372207	Yes, Yes
Tested model A	{ARMA}(1,1) + D_{2020}	{GARCH}(1,1)	p=0.0411846	p=0.253298	No, Yes
Tested model B…	{ARMA}(0,0) + D_{2020}	{GARCH}(1,1)	p=0.0000	p=0.0000	No, No

Table 2

$$\{AR\}(1) + D_{2013} + D_{2020} - GARCH(1,1). \quad || \quad \{AR\}(1) + D_{2020} - GARCH(1,1)$$

```
ARIMAX(1,0,0) Model (Gaussian Distribution)

 Effective Sample Size: 3276
 Number of Estimated Parameters: 7
 LogLikelihood: 13215.7
 AIC: -26417.5
 BIC: -26374.8

                 Value       StandardError    TStatistic      PValue
                _________    _____________    __________    __________

 Constant       0.0001202     0.00011648          1.032       0.30207
 AR{1}          -0.097883      0.017578          -5.5685     2.569e-08
 Beta(1)      -4.4051e-05     0.00014928        -0.29509       0.76793
 Beta(2)       2.1188e-05     0.00016802         0.12611       0.89965


GARCH(1,1) Conditional Variance Model (Gaussian Distribution)

                 Value       StandardError    TStatistic      PValue
                _________    _____________    __________    __________

 Constant      3.9525e-07     1.9075e-07         2.072       0.038262
 GARCH{1}        0.89005      0.0062708        141.94              0
 ARCH{1}         0.09704      0.0069558        13.951      3.1043e-44
```

```
ARIMAX(1,0,0) Model (Gaussian Distribution)

 Effective Sample Size: 3276
 Number of Estimated Parameters: 6
 LogLikelihood: 13215
 AIC: -26418
 BIC: -26381.5

                 Value       StandardError    TStatistic      PValue
                _________    _____________    __________    __________

 Constant      9.809e-05     7.2109e-05         1.3603       0.17374
 AR{1}         -0.092953      0.017237         -5.3925     6.9477e-08
 Beta(1)      2.2755e-05     0.00015634        0.14555       0.88428


GARCH(1,1) Conditional Variance Model (Gaussian Distribution)

                 Value       StandardError    TStatistic      PValue
                _________    _____________    __________    __________

 Constant      3.6611e-07     1.8445e-07         1.9848       0.047163
 GARCH{1}        0.89544      0.0058783        152.33              0
 ARCH{1}        0.091783      0.0064145        14.309      1.9267e-46
```

Figure 5

Comparing BIC of both these adequate models(interms of both mean and variance), we select the smallest BIC. Since the **BIC imposes a stronger penalty on complexity**, this criterion favors the **most parsimonious model**. Based on this principle, the **Single Covid Dummy specification** is selected as the superior **Benchmark model** . Although the coefficient for the Covid Dummy variable was found to be statistically insignificant, the model remains the most parsimonious and adequate specification to pursue for out-of-sample forecasting.

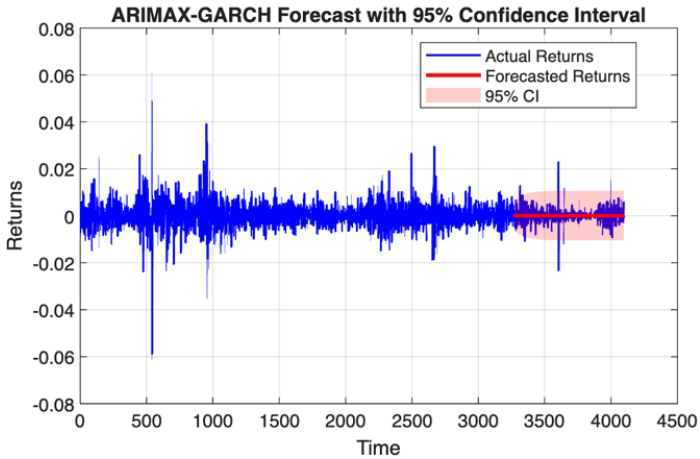

Figure 6

The RMSE of the Benchmark model is 0.002689.

4.3. Egarch models

After testing and ARIMAX(1,0,0) mean model combined with : EGARCH(1,1), EGARCH(1,2), EGARCH(2,1), EGARCH(1,3), EGARCH(3,1), EGARCH(2,3), EGARCH(3,2), The most parsimonious adequate model according to the Lju jung Box test was **ARIMAX-EGARCH(3,3)** (Appendix 1).

The **ARIMAX-EGARCH(3,3)** returns a reliable forecast (Appendix 2) with a RMSE of. 0.002693.

Then after checking the **ARIMAX-EGARCH (1,1) -X(VIX)** model results (Appendix 1), the model successfully passes the Ljung Box test . The model provided a deeper look into the nature of the volatility shocks with statistically significant leverage parameter(Appendix 1). After the forecast (Appendix 2) the RMSE is equal to 0.002690.

4.4. Model Selection based on Forecast Accuracy

The final selection criteria are based on **out-of-sample forecasting performance**, using the Root Mean Squared Forecast Error:

Model	In-Sample Fit (BIC)	Out-of-Sample Accuracy (RMSE)	Conclusion on Accuracy
Benchmark - ARIMAX-GARCH(1,1)	-26381.5	**0.002689**	Benchmark Forecast
ARIMAX-EGARCH(3,3)	-26358.4	0.002693	Alternative Forecast
ARIMAX-EGARCH(1,1)-X(VIX)	-26394.8	0.002690	Alternative Forecast

Applying the principle of **Parsimony**, the simplest adequate model that performs equally well in forecasting is preferred. The model was identified as the superior asymmetric specification, having the lowest **BIC**among the tested models. However, since its **Root Mean Squared Forecast Error** was nearly identical to that of the simpler Benchmark model, the model's added complexity did not yield superior out-of-sample accuracy. Therefore, the ARIMAX-GARCH(1,1) **model is conclusively selected as the final, most robust Benchmark Forecasting Model.**

5. Conclusion:

5.1. Summary of Findings

This study successfully applied Garch-family models to the daily logarithmic returns of the exchange rate, effectively modeling its time-varying volatility. The mean equation used in the models was an ARIMAX(1,0,0).

1. **Structural Breaks:** The Chow test confirmed statistically significant shifts in the mean return dynamics following the 2013 Taper Tantrum and the 2020 COVID-19 shock, validating the necessity of including structural break dummy variables. For parsimony and no loss of information only the Covid dummy is used in the selected benchmark model.
2. **Volatility Dynamics:** The currency pair exhibits **pronounced volatility clustering** and **strong persistence**, indicated by GARCH coefficients near unity.
3. **Asymmetry:** The estimation provided empirical evidence for the **Leverage Effect** (lambda parameter), confirming that negative shocks (Rupee depreciation) amplify future volatility more than positive shocks of equal magnitude.

5.2. Model Selection and Implication

While the ARIMAX-EGARCH-x(VIX) model provides a superior theoretical description of the volatility process by capturing asymmetry, the out-of-sample forecast evaluation using the demonstrated that its added complexity does not result in superior **mean return forecast accuracy**.

Therefore, adhering to the principle of parsimony and prioritizing predictive efficiency, the ARIMAX-GARCH(1,1) model is conclusively selected as the most appropriate and robust **Benchmark Model** for generating short-term conditional mean forecasts for the exchange rate. This model provides a reliable framework for risk management applications.

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Appendix1:

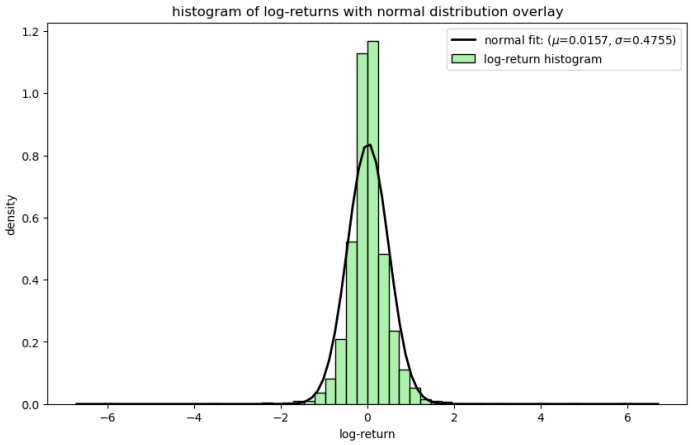


figure 1

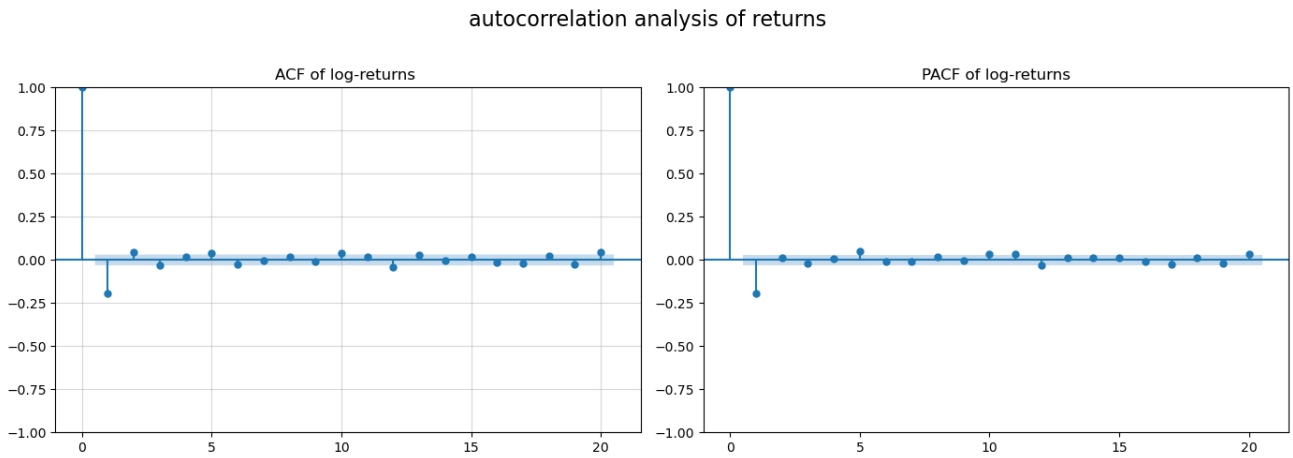


figure 2

ARIMAX-EGARCH (3,3)

ARIMAX(1,0,0) Model (Gaussian Distribution)

```
Effective Sample Size: 3276
Number of Estimated Parameters: 13
LogLikelihood: 13231.8
AIC: -26437.6
BIC: -26358.4
```

	Value	StandardError	TStatistic	PValue
Constant	0.00016106	7.7569e-05	2.0763	0.037869
AR{1}	-0.083408	0.018847	-4.4255	9.6239e-06
Beta(1)	0.00014305	0.00014424	0.99172	0.32134

EGARCH(3,3) Conditional Variance Model (Gaussian Distribution)

Selam Mahmud Ali
48015A

	Value	StandardError	TStatistic	PValue
Constant	-0.059942	0.01535	-3.9051	9.4195e-05
GARCH{1}	0.99907	0.39474	2.531	0.011375
GARCH{2}	0.060504	0.61183	0.098891	0.92122
GARCH{3}	-0.065444	0.23543	-0.27797	0.78103
ARCH{1}	0.22084	0.025433	8.6829	3.8585e-18
ARCH{2}	-0.018426	0.09555	-0.19285	0.84708
ARCH{3}	-0.11567	0.073189	-1.5805	0.114
Leverage{1}	0.11257	0.019422	5.7963	6.7806e-09
Leverage{2}	-0.064882	0.048975	-1.3248	0.18524
Leverage{3}	-0.013262	0.042931	-0.30892	0.75738

ARIMAX EGARCH -X:

ARIMAX(1,0,0) Model (Gaussian Distribution)

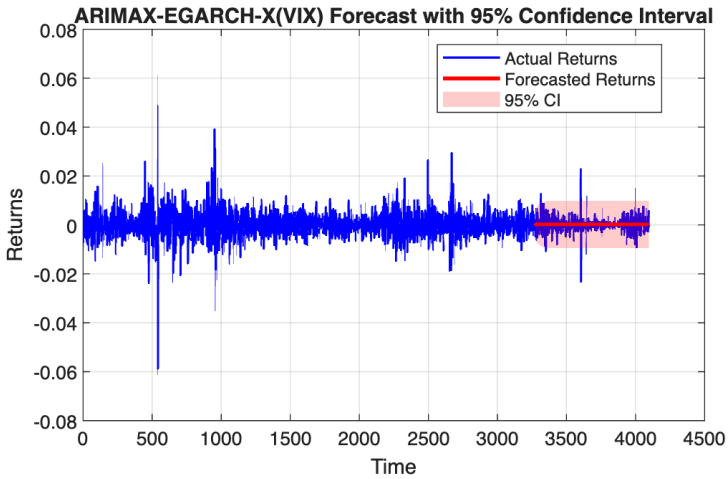
Effective Sample Size: 3276
Number of Estimated Parameters: 7
LogLikelihood: 13225.7
AIC: -26437.5
BIC: -26394.8

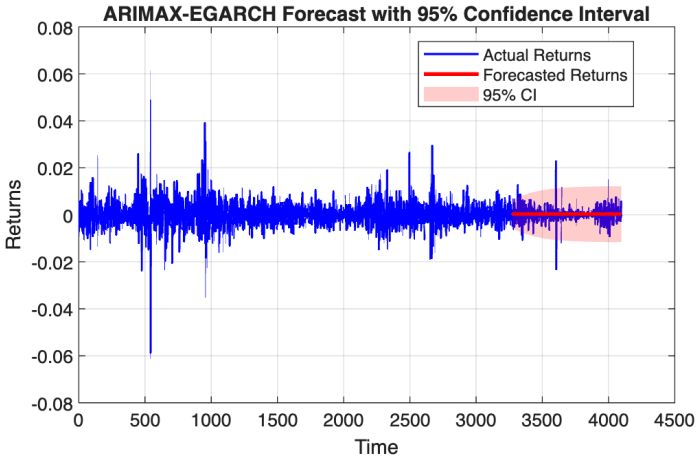
	Value	StandardError	TStatistic	PValue
Constant	0.00018685	8.0938e-05	2.3086	0.020965
AR{1}	-0.075117	0.017964	-4.1814	2.8971e-05
Beta(1)	3.6362e-05	0.00014597	0.24911	0.80328

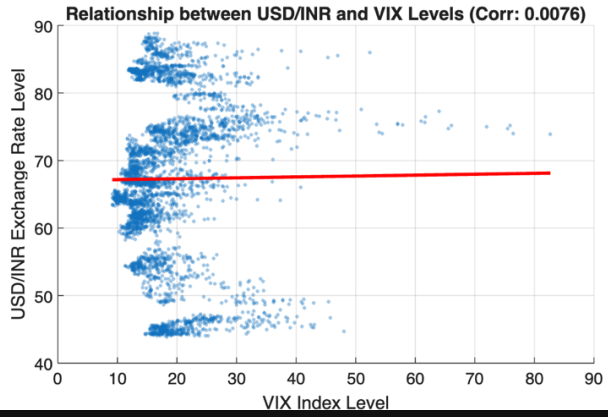
EGARCH(1,1) Conditional Variance Model (Gaussian Distribution)

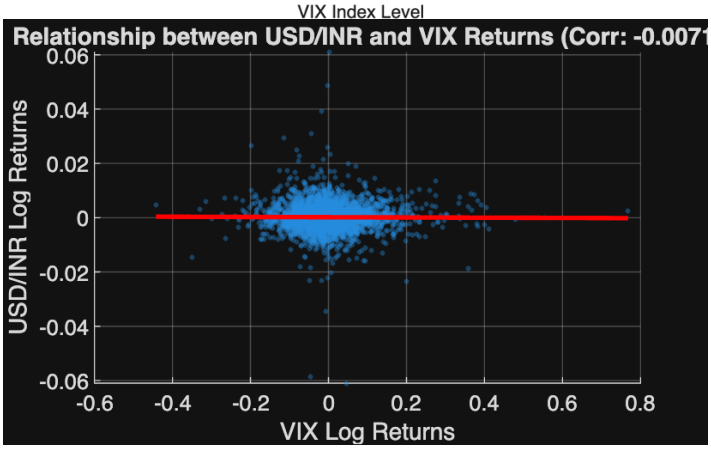
	Value	StandardError	TStatistic	PValue
Constant	-0.40194	0.036141	-11.121	9.8707e-29
GARCH{1}	0.96217	0.003368	285.68	0
ARCH{1}	0.24441	0.012491	19.566	3.0009e-85
Leverage{1}	0.041952	0.005114	8.2032	2.34e-16

Appendix 2:









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